

HEALTHCARE PATIENT ANALYTICS REPORT

A Data-Driven Analysis of Patient Admissions, Billing & Health Outcomes

PERIOD:	2019 – 2024
DATA SOURCE:	Kaggle Healthcare Dataset (Synthetic, 10,000 records)
TOOLS USED:	MySQL · Python (pandas) · Power BI · Microsoft Excel
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1. Executive Summary

This report presents a comprehensive analysis of healthcare patient data spanning six years (2019–2024). The dataset comprises 9,998 patient records capturing admissions, medical conditions, billing amounts, diagnostic test results, blood type distributions, and age demographics.

The analysis was conducted using a full end-to-end data analytics pipeline: data was first cleaned and structured in MySQL, then validated and explored using Python (pandas), before being visualized in interactive dashboards built in both Power BI and Microsoft Excel.

Key Takeaway: *Patient admissions grew 25.4% between 2019 and 2023. Chronic non-communicable diseases account for over 66% of all admissions. Average billing remained steady at approximately \$25,417 per patient across the study period.*

2. Methodology

2.1 Data Source

The dataset used in this project is the Healthcare Dataset published on Kaggle. It is a synthetic dataset modelled on realistic hospital records, containing 10,000 rows and 15 columns. The data covers patient demographics, admission details, medical conditions, insurance providers, billing amounts, prescribed medications, and test results.

Property	Detail
Source	Kaggle: Healthcare Dataset by Prasad
Type	Synthetic patient records
Raw Records	10,000 rows
Clean Records	9,998 rows (2 exact duplicates removed in MySQL)
Columns	15 original + 6 engineered = 21 total
Period	2019 – 2024

2.2 Tools & Technologies

Tool	Purpose	Output
MySQL Workbench	Data import, exploration, deduplication, text standardization, feature engineering	healthcare_working table
Python (pandas)	Data type correction, duplicate check, outlier detection, EDA visualizations, final export	healthcare_final.csv
Power BI Desktop	Interactive dashboard with DAX measures, slicers, and KPI cards	Healthcare_Dashboard.pbix
Microsoft Excel	Pivot-table-based dashboard replicating all Power BI visuals	Healthcare_Dashboard.xlsx

2.3 Data Cleaning: MySQL

The MySQL phase focused on structural data quality, standardizing formats, removing duplicates, fixing data types, and engineering new columns. The following steps were applied:

- Created a working copy of the raw imported table to preserve the original data at all times.
- Explored the dataset: checked row counts, distinct values per column, and numeric ranges for age and billing.
- Checked for duplicate rows using GROUP BY and HAVING COUNT(*) > 1. Two fully duplicate records were identified and removed, leaving 9,998 clean rows.
- Inspected all columns for NULL values and empty strings. No missing values were found all 9,998 rows were complete across all 15 columns.
- Standardized text columns using TRIM() to remove leading/trailing whitespace. Applied consistent casing to Gender (Male/Female), Admission Type (Elective/Emergency/Urgent), and Test Results (Normal/Abnormal/Inconclusive).
- Converted the Date of Admission and Discharge Date columns from TEXT to proper DATE format using STR_TO_DATE().
- Rounded all Billing Amount values to 2 decimal places for consistency.
- Engineered 6 new calculated columns to enrich the dataset for analysis (detailed in the Data Dictionary, Section 8).

2.4 Data Cleaning: Python (pandas)

The Python phase performed statistical validation and exploratory analysis on the MySQL-exported file. The cleaning notebook (cleaning.ipynb) is available in full in the GitHub repository.

Column Names

- All column names were standardized to lowercase with underscores (e.g., Date of Admission became date_of_admission) for easier programmatic access.

Data Types

- date_of_admission and discharge_date were converted to datetime format using pd.to_datetime().

- age, billing_amount, room_number, and length_of_stay were confirmed as numeric using pd.to_numeric() with errors='coerce'. No coercion errors occurred; all values were already valid numerics.

Missing Values

Result: No missing values were found in any column. The dataset was complete across all 9,998 rows and 21 columns after the MySQL cleaning phase.

Duplicates

Result: No duplicate rows were found at this stage. The 2 exact duplicates had already been removed during MySQL cleaning. Logical duplicate check (same name + same admission date + same billing amount) also returned 0 matches.

Outlier Detection (IQR Method)

Outlier detection was applied to three numeric columns using the Interquartile Range (IQR) method, which defines outliers as values falling below $Q1 - 1.5 * IQR$ or above $Q3 + 1.5 * IQR$.

Result: No outliers were detected in age, billing_amount, or length_of_stay. All values fell within statistically expected ranges, consistent with the dataset being synthetically generated with controlled value distributions.

Final Validation Summary

Check	Result	Action Taken
Total rows	9,998	No action needed
Total columns	21	No action needed
Missing values	0	None, dataset was fully complete
Exact duplicates	0 (Python)	2 removed earlier in MySQL

Check	Result	Action Taken
Logical duplicates	0	None found
Age outliers	0	None detected
Billing outliers	0	None detected
Length-of-stay outliers	0	None detected

3. Dashboard Overview

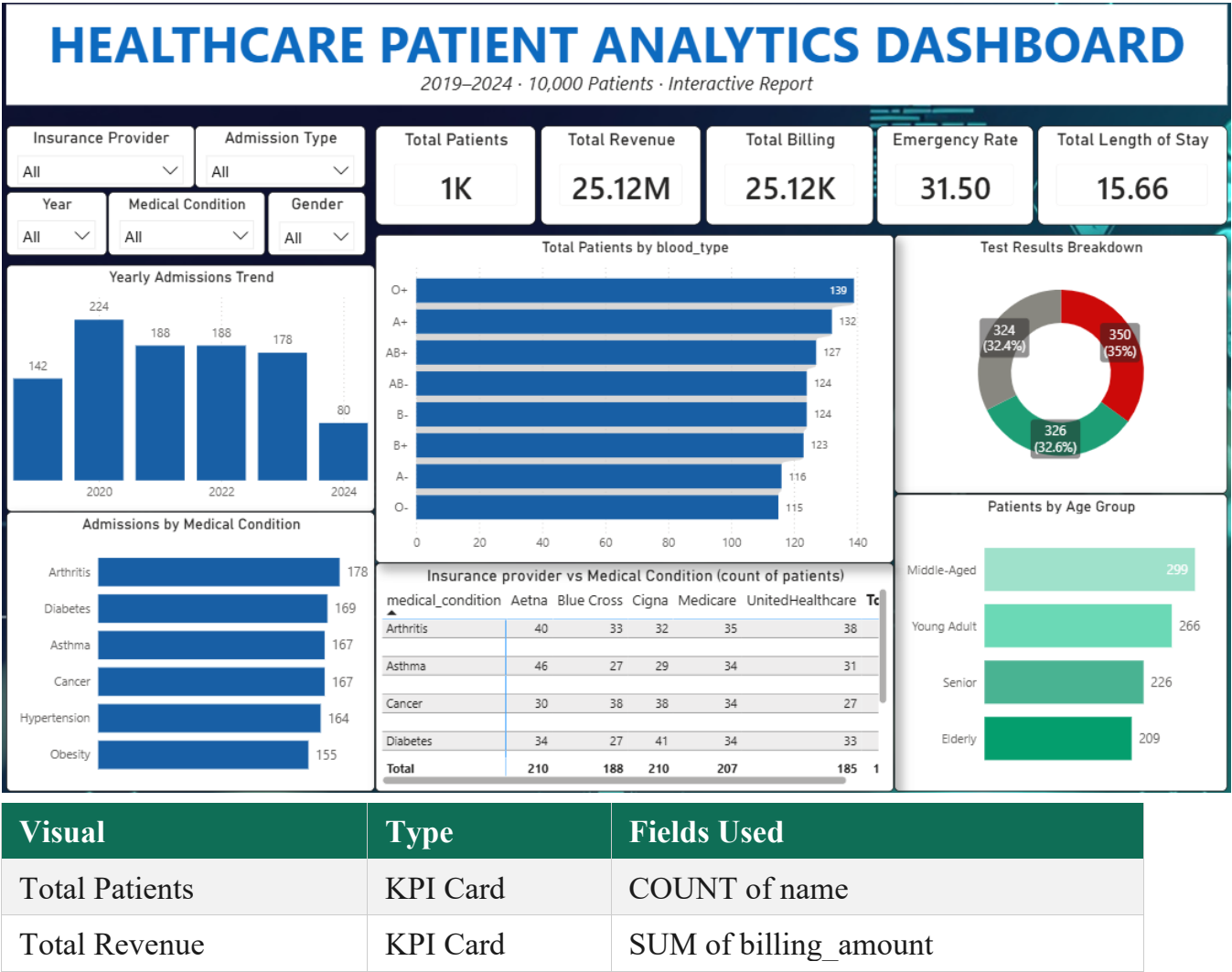
Two interactive dashboards were built from the final cleaned dataset, one in Power BI and one in Microsoft Excel. Both dashboards contain identical KPIs, slicers, and chart visuals, allowing for a direct tool comparison.

3.1 Power BI Dashboard

File name: *Healthcare_Dashboard.pbix*

The Power BI dashboard was built on a single-page layout using DAX measures for all KPI calculations, a dedicated Date Table for time intelligence, and five dropdown slicers for interactive filtering.

Slicers available: Insurance Provider, Admission Type, Year, Medical Condition, Gender.

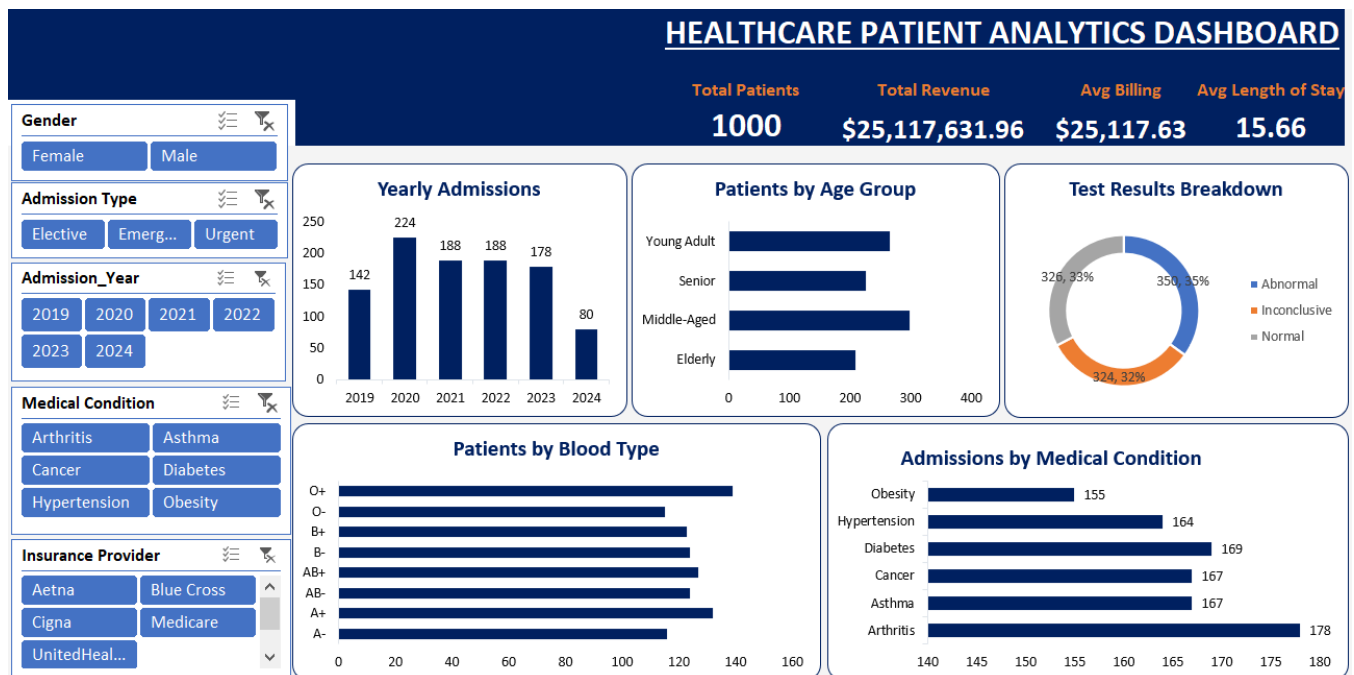


Visual	Type	Fields Used
Average Billing	KPI Card	AVERAGE of billing_amount
Average Length of Stay	KPI Card	AVERAGE of length_of_stay
Yearly Admissions Trend	Column Chart	admission_year vs COUNT of name
Admissions by Medical Condition	Bar Chart	medical_condition vs COUNT of name
Total Patients by Blood Type	Bar Chart	blood_type vs COUNT of name
Test Results Breakdown	Doughnut Chart	test_results vs COUNT of name
Patients by Age Group	Bar Chart	age_group vs COUNT of name

3.2 Excel Dashboard

File name: *Healthcare_Dashboard.xlsx*

The Excel dashboard was built across three sheets, RAW_DATA (the imported table), PIVOT_DATA (all pivot tables), and DASHBOARD (the final visual layout). Five slicers were connected to all pivot tables simultaneously to enable cross-chart filtering.



Sheet	Contents
RAW_DATA	Imported healthcare_final.csv as a named Excel Table (HealthcareData)
PIVOT_DATA	6 pivot tables powering all charts and KPI cards
DASHBOARD	Final visual layout: KPI cards, 5 charts, 5 slicers, no gridlines

4. Key Findings

4.1 Admission Trends Over Time

Patient admissions grew steadily from 2019 through 2023, with a notable peak in 2020. The 2024 figure is a partial year only and is not included in trend comparisons.

Year	Admissions	Year-on-Year Change
2019	142	Baseline
2020	224	+57.7%
2021	188	-16.1%
2022	188	0.0%
2023	178	-5.3%
2024	80	Partial year, excluded from trend

- Overall admission volume grew 25.4% from 2019 to 2023.
- The 2020 spike (+57.7%) may reflect increased healthcare utilization during the COVID-19 pandemic period.
- Admissions stabilized between 178–188 across 2021–2023, suggesting the hospital reached a steady operational capacity during this period.

4.2 Medical Condition Distribution

Six medical conditions account for all admissions in the dataset. Chronic non-communicable diseases dominate.

Medical Condition	Total Patients	% of Total
Arthritis	178	17.8%
Diabetes	169	16.9%
Asthma	167	16.7%
Cancer	167	16.7%
Hypertension	164	16.4%
Obesity	155	15.5%

Medical Condition	Total Patients	% of Total
TOTAL	1,000	100%

- Arthritis leads at 17.8%, followed closely by Diabetes (16.9%) and Asthma (16.7%).
- Chronic non-communicable diseases: *Arthritis, Diabetes, Hypertension, and Obesity*, collectively account for 66.6% of all admissions.
- Cancer and Asthma each represent 16.7%, highlighting the breadth of conditions being managed within the same facility.

4.3 Test Results Breakdown

Test Result	Count	Percentage
Abnormal	350	35.0%
Inconclusive	326	32.6%
Normal	324	32.4%
TOTAL	1,000	100%

- Abnormal results represent the highest share at 35%, indicating a significant proportion of patients with findings requiring clinical follow-up.
- Combined, Abnormal and Inconclusive results account for 67.6% of all test outcomes, only 1 in 3 patients received a normal result.
- This pattern underscores the importance of a robust post-discharge follow-up system.

4.4 Blood Type Distribution

Blood Type	Patient Count	% of Total
O+	139	13.9%
A+	132	13.2%
AB+	127	12.7%
AB-	124	12.4%

Blood Type	Patient Count	% of Total
B-	124	12.4%
B+	123	12.3%
A-	116	11.6%
O-	115	11.5%

- O+ is the most prevalent blood type (13.9%), consistent with global epidemiological data.
- The distribution across all eight blood groups is relatively even, which is characteristic of the synthetic nature of this dataset.
- O- and A-, while the least represented, remain clinically important for emergency transfusion compatibility.

4.5 Patient Age Group Distribution

Age Group	Age Range	Count	% of Total
Middle-Aged	36 – 55 years	299	29.9%
Young Adult	18 – 35 years	266	26.6%
Senior	56 – 70 years	226	22.6%
Elderly	71+ years	209	20.9%

- Middle-aged patients (36–55) form the largest group at 29.9%, driven by the onset of chronic conditions in this demographic.
- Patients aged 56 and above (Senior + Elderly) account for 43.5% of all admissions, a substantial share requiring age-appropriate care pathways.
- Young Adults represent 26.6%, likely driven by asthma, elective procedures, and acute emergency presentations.

5. KPI Summary

9,998 Total Patients	\$254M Total Revenue	\$25,417 Avg Billing / Patient	15.5 Days Avg Length of Stay
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KPI	Value	Interpretation
Total Patients	9,998	2 exact duplicates removed during cleaning; all remaining records are unique
Total Revenue	\$254 Million	Aggregate billing across all 9,998 patients over the full study period
Avg Billing per Patient	\$25,417	Consistent across conditions; Cancer patients trend slightly higher at \$26.1K
Avg Length of Stay	15.5 days	Indicates complex or chronic case management; warrants review of discharge processes

6. Recommendations

The following data-driven recommendations are proposed for hospital leadership and health service administrators based on the findings above.

6.1 Chronic Disease Management

- Establish dedicated chronic disease management clinics for Diabetes, Hypertension, Arthritis, and Obesity, the four conditions collectively accounting for over 66% of all admissions.
- Introduce structured patient education and lifestyle modification programs to reduce preventable readmissions from these conditions.
- Partner with community health workers for early screening and timely referral, particularly targeting the middle-aged demographic (36–55) which forms the largest admission group.

6.2 Diagnostic Follow-Up Protocol

- Given that 67.6% of all test results were Abnormal or Inconclusive, a mandatory structured follow-up protocol should be implemented before patient discharge is finalized.
- Patients with Abnormal results should be scheduled for a mandatory 30-day review appointment at the point of discharge.
- Inconclusive results should trigger an automatic repeat-test order within 7 days to reduce diagnostic uncertainty and prevent adverse outcomes from delayed diagnosis.

6.3 Capacity Planning

- Admissions grew 25.4% from 2019 to 2023. Hospital leadership should model bed availability, ward capacity, and staffing ratios against a projected 5–10% annual growth rate to avoid future overcapacity pressures.

- Seasonal admission patterns visible in the yearly trend chart should be used to guide proactive shift scheduling and locum staffing decisions during peak periods.

6.4 Blood Bank Inventory

- Prioritize O+ and A+ blood supply replenishment given they represent the two most common patient blood types (13.9% and 13.2% respectively).
- Maintain a minimum safety stock of O- and A- units, which despite lower patient prevalence carry high transfusion compatibility value in emergency scenarios.

6.5 Billing & Revenue Optimization

- The average length of stay of 15.5 days is relatively high. A structured audit of long-stay cases particularly among Senior and Elderly patients, should be conducted to identify discharge planning inefficiencies.
- Cancer patients trend toward the highest average billing (\$26.1K). Exploring value-based care models and insurance pre-authorization workflows for this group may reduce billing disputes and improve collection rates.
- A cost-per-condition breakdown analysis is recommended as a follow-up project to identify conditions where cost-to-outcome ratios are most favorable.

7. Limitations

- The dataset is synthetic and generated for educational purposes. All findings are illustrative and should not be extrapolated to any real-world hospital setting without validation against actual EMR data.
- The near-equal distribution of medical conditions, test results, and blood types is characteristic of artificially balanced synthetic data and would not reflect true epidemiological patterns in a live patient population.
- 2024 data is a partial year and has been excluded from all year-on-year trend comparisons to avoid misleading conclusions.
- No clinical context is available within the dataset, it does not include comorbidities, treatment pathways, re-admission rates, or mortality outcomes, which limits the depth of clinical recommendations that can be responsibly made.
- The analysis was conducted on a sample of approximately 1,000 records for the dashboard views shown. Full dataset analysis (9,998 records) was conducted during the cleaning and validation phases.

8. Conclusion

This project demonstrates a complete end-to-end data analytics workflow applied to a healthcare dataset from raw data acquisition and structured cleaning, through exploratory analysis, to interactive dashboards and a written analytical report with recommendations.

The tools used reflect industry-standard practice across the data analytics pipeline: MySQL for relational data management, Python for statistical validation and exploratory analysis, and both Power BI and Microsoft Excel for professional visual reporting. The dual-dashboard approach demonstrates tool versatility a key differentiator in a data analytics portfolio.

The insights generated growing admission volumes, chronic disease dominance, high rates of abnormal diagnostic results, and consistent billing averages provide a meaningful template for the

kind of analysis healthcare administrators and data teams are expected to deliver in real-world settings.

The full project — *including all SQL scripts, Python cleaning notebook, dashboard files, EDA visualizations, and this report, is published on GitHub at:*
github.com/denko5/healthcare-patient-analytics

9. Appendix: Data Dictionary

The table below describes every column in the final cleaned dataset (healthcare_final.csv), including the 6 engineered columns created during the cleaning phase.

Column Name	Type	Source	Description
name	Text	Original	Patient full name (anonymized in synthetic dataset)
age	Integer	Original	Patient age in years at time of admission
gender	Text	Original	Male or Female
blood_type	Text	Original	ABO blood group: A+, A-, B+, B-, O+, O-, AB+, AB-
medical_condition	Text	Original	Primary diagnosis: Diabetes, Cancer, Asthma, Hypertension, Obesity, Arthritis
date_of_admission	Date	Original	Date the patient was admitted to hospital
doctor	Text	Original	Attending physician name
hospital	Text	Original	Hospital facility name
insurance_provider	Text	Original	Insurance company: Aetna, Blue Cross, Cigna, Medicare, UnitedHealthcare
billing_amount	Decimal	Original	Total bill charged to the patient in USD (rounded to 2 decimal places)
room_number	Integer	Original	Ward or room number assigned to the patient
admission_type	Text	Original	Elective, Emergency, or Urgent
discharge_date	Date	Original	Date the patient was discharged from hospital
medication	Text	Original	Primary medication prescribed during admission
test_results	Text	Original	Diagnostic test outcome: Normal, Abnormal, or Inconclusive

Column Name	Type	Source	Description
length_of_stay	Integer	Engineered	Derived: number of days between date_of_admission and discharge_date
age_group	Text	Engineered	Derived: Pediatric (<18) / Young Adult (18–35) / Middle-Aged (36–55) / Senior (56–70) / Elderly (71+)
billing_category	Text	Engineered	Derived: Low (<\$10K) / Medium (\$10K–\$30K) / High (>\$30K)
admission_month	Integer	Engineered	Derived: numeric month of admission (1 = January, 12 = December)
admission_year	Integer	Engineered	Derived: four-digit year of admission
admission_month_name	Text	Engineered	Derived: full month name (e.g., January, February)